SESSION-1

1. Name 5 IT job roles and write one line about what each role does.
   Software developer , web developer , aiml engineer , devops engineer , dba .

2. List 3 product companies and 3 service companies with one example of work each does.
   - 3 product companies
     - Google- develop the google crome.
     - Microsoft- develop Microsoft application.
     - Amazon
     - Flipcart
   - 3 service based company
     - Tcs
     - Wipro
     - Infosys.

3. Write the full forms and one-line use of: Github,Slack,jira,ide,Api.
   - Github : github is cloud based platform and ai powered developer tool used to store, share, and collaborate on code.
   - Slack : Searchable log of All Conversation and knowledge. Used a centralized messaging platform for team communication and collaborate.
   - Jira : (Derived from Gojira, the Japanese name for Godzilla). Used to track issues, manage project workflows,and plan agile sprints.
   - IDE : Integrated Development Environment. Used as a software application that provides tools to write and test code.
   - API : Application Program Interface. Used to allow different software systems to communicate and share data with one another.

4. Name any 5 programming languages and mention one popular use case for each.
   - Java
   - Python
   - C++
   - .net
   - Sql
5. What is the difference between a developer and tester ? Answer in 2 lines.
   A developer is responsible for writing the code and building software applications or features . A tester evaluates that software to identify bugs ,ensuring it functions currently and meets quality standards before release.

SESSION-2

1. What is the three main layers of software architecture? Write one line about each.

    - Presentation Layer : the presentation layer is also called the UI layer , handles the Interections that users have with software. It is the most visible layer.
    - Application Layer: the application layer handles the main program of architecture.
    - Database Layer: The Database layer is where the system stores all the data. Search , Insert , update and delete operations occur here frequently.

2. Define client and server in your own words with one real-life example.

    Imagine a customer sitting at a restaurant. He is waiting for the server to come by and take his order.

3. Write 2 difference each between desktop apps. Web apps,and mobile apps.
    Desktop apps: must be installed directly onto a computer local driver to function.
    Web apps: no installation is required they are accessed via web browsers.
    Mobile apps: downloaded and installed onto smartphones or tablets. Ex.Google play

4. Name any 3 databases and mention whether each is SQL or NoSql.
    - MySQL - SQL
    - MongoDB- No SQL
    - PostgreSQL-SQL

SESSION-3

1. Write the full form and one-line use of:HTTP,HTTPS,FTP,SMTP.

   HTTP: hyper text transfer protocol. To transfer data like html , pages,images and videos, between a client and a server.

   HTTPS: hyper text transfer protocol secure. Entering your web address with the secure protocol In the address bar, or configuring a command in your server's settings to force it.

   FTP: file transfer protocol. ftp is a standard set of rules used to transfer computer files between two computers or devices connected over a network.

   SMTP: simple mail transfer protocol. Smtp is an internet standard used to send and route outgoing emails between servers.

2. What is the difference between IPv4 and IPv6? answer in 2 lines.
   Ipv4: ipv4 address size 32 bit.and address format dotted decimal. This is manual configuration.must support DHCP or be configured manually.
   Ipv6: Ipv6 address size 128 bit . and address format hexadecimal notaions. Does not required DHCP or manual configuration. It supports stateless auto configuration .

3. Run nslookup google.com on your system and write down the Ip address you get.

   Server: UnKnown

   Address: 10.231.165.47

   Non-authoritative answer:

   Name: google.com

   Addresses: 2404:6800:4000:100e::65

   2404:6800:4000:100e::8b

   2404:6800:4000:100e::71

   2404:6800:4000:100e::6 142.250.76.174

4. What do these http codes mean: 200,301,404,500,403?
    - 200: Ok – success
    - 301: moved permanenty- redirect
    - 404: not found- page missing
    - 500 : internal server error- server error
    - 403: forbidden – access denide.

5. Explain DNS hierarchy(root,TLD,domain,subdomain)with google.com as an example.

    - Root level – (.)-the ultimate apex starting point of the entire internet namespace .
    - TLD: top level domain. (com)- categorize the domain type (eg. Commercial ,educational ,country-specific).
    - Second level domain: (Google)- the unique ,registered brand or organization identity.
    - Subdomain: (www)- A specific subsection ,server,or host designated by the domain owner.

SESSION-4

1. List all 6 phases of SDLC in order with one line about each.

    1. Requirement Gathering: Defining the project scope, objectives, resource requirements, and estimating costs and timelines.
    2. Analysis : Gathering and documenting exact user and system needs, often compiled into a Software Requirement Specification (SRS) document.
    3. Designing: Transforming requirements into software architecture, including user interfaces, databases, and system security protocols.
    4. Implementation: Translating the design specifications into functional code and building the actual software product.
    5. Testing : Executing the software to identify and fix any bugs, ensuring the product meets quality standards and initial requirements.
    6. Maintenance: Providing continuous support, fixing post-launch issues, and updating the software with new features as needed.

2. Write 3 difference between agile and waterfalls model.
    - Agile model: work is divided into short cycle to deliver features quickly and continuously.
      Continuous customer involvement ensure better alignment with changing requirements.
      Chages can be easily incorporated at any stage of development.
    - Water Fall model : Development follows fixed phases such as requirement → design → development → testing → deployment.
      Customer involvement is limited mainly to the intial and final stages .
      Changes are difficult and costly once a phase is completed.

3. Name the 3 main scrum roles and write one line about the responsibility of each.
    - Product Owner: maximizes the value of the product by manging and ordering the product backlog.
    - Scrum master: ensures the scrum team adheres to scrum theory ,practices and rules.
    - Developers: the professionals who do the work of delivering a usable increment of the product at the end of each sprint.

4. What is a sprint in scrum ? Mention its typical duration.

A spring scrum is a fixed -length , time boxed period where a team work to complete a set amount of work .

- Typical Duration:

  Official guideline: according to official scrum guide sprints must be one month or less in duration to maintain consistency.

  Real World Practice: the most common and industry – standard duration in 2 week. However teams can choose any consistent duration ranging for 1 to 4 weeks .

  Consistency: once a sprint length in chosen , it must remain the same for all consecutive sprints.

5. What do these Jira board column mean: To Do, In Progress , Done ,Backlog ?

   Jira board column represent the sequential steps of your team's workflow .they visualize the journey of a task from an idea to completion, categorized as follow.

   - ToDo: the staging area for the current sprint or work cycle . Tasks moved here are ready to be started ,but no one is actively working on them yet.
   - In Progress: work that team members have actively started. It signifies that someone has claimed the task and is currently executing it.
   - Done: Completed tasks that meet your team's requirements. In Jira, this is typically the right-most column; moving a task here signals the end of its workflow.
   - Backlog: The holding area for all upcoming work, ideas, and feature requests. Tasks here are unassigned and unprioritized. This is where product managers and teams organize what they will build next.

SESSION-5

1. Write the git command for each: initialize a repo ,stage all files commit with a message push to remote ,pull latest changes.

   The core commands

   - Initialize a repo: git init
   - Stage all files :git add .(or git add -A)
   - Commit with message :git commit -m “your message here”
   - Push to remote : git push origin main(replace main with your specific branch name)
   - Pull latest change :git pull

      1. Setting up your project : to start version tracking a brand new folder navigate to your root directory in the terminal an initialize it.

         Bash : git init
      2. Preparing your changes : Before saving your progress, you must tell Git which files to bundle into your snapshot. You can prepare every single changed and new file at once.

         Bash : git add.
      3. Saving your progress locally : create a permanent save point in your local history with a clear , descriptive message.

         Bash : git commit -m”add initial project files”
      4. Uploading to remote server: to share your local commits with a platform like github or gitlab, use the push command.

         Bash : git push origine main
      5. Fetching team updates: to fetch updates from your team and instantly merge them directly into your current working branch , use.

         Bash : git pull

2. What is the difference between git fetch and git pull ? answer in 2 lines.

   Git fetch: downloads new data from a remote repository.

Git fetch formula is git fetch .

Working directory is unchanged safe to run anytime.

This is a clean and safe low risk level.

Git pull: downloads data and integrates it into local files.

Gil pull formula git fetch+ git merge.

Git pull Working directory modified instantly.

3. Create a new github repository ,push any small text file from your local system, and share the repo link.

   Github-> “new repository”

   Git init

   Git add .

   Git status

   Git commit -m “first commit”

   Git branch -m main

   Git remote add origin https://github.com/vikashpandey770/Software-Eng/blob/main/hekj.txt

   Git push -u origin main

4. What is the difference between a fork and pull Request? Answer in 2 line.

   Fork: fork is a personal copy of someone else's repository.
   Fork visibility changes are only visible in your forked copy until shared.
   Pull Request: pull request to merge code changes into a main project.
   Pull request visibility changes are public in the PR for discussion and review.

5. Create a new branch named feature -test,make one commit on it, and merge it Back to the main branch . write the commands you used.

   # 1. Create and switch to the new branch named 'feature-test'

   git checkout -b feature-test

   # 2. Stage your changes (after making edits to your project files)

   git add

   # 3. Create a commit on the 'feature-test' branch

   git commit -m "Add test feature modifications

   # 4. Switch back to the main branch

   git checkout main

   # 5. Merge the 'feature-test' branch into main

   git merge feature-test